EFFICACY AND SAFETY OF PUSH-DOSE NOREPINEPHRINE FOR INTRAOPERATIVE HYPOTENSION IN NON-OBSTETRIC SURGERY: A SYSTEMATIC REVIEW

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BACKGROUND

Intraoperative hypotension (IOH), defined as MAP <65 mmHg, occurs in up to 30% of non-cardiac surgeries and is independently associated with myocardial injury, acute kidney injury, and 30-day mortality. Push-dose (bolus) norepinephrine (NE) is gaining popularity; however, high-quality evidence remains limited.

OBJECTIVE

To systematically evaluate efficacy and safety of push-dose NE for IOH in non-obstetric surgery vs other vasopressors, following PRISMA 2020 guidelines with GRADE certainty assessment.

METHODS

Design: Systematic Review · PRISMA 2020 compliant

Databases: PubMed · Scopus · ProQuest

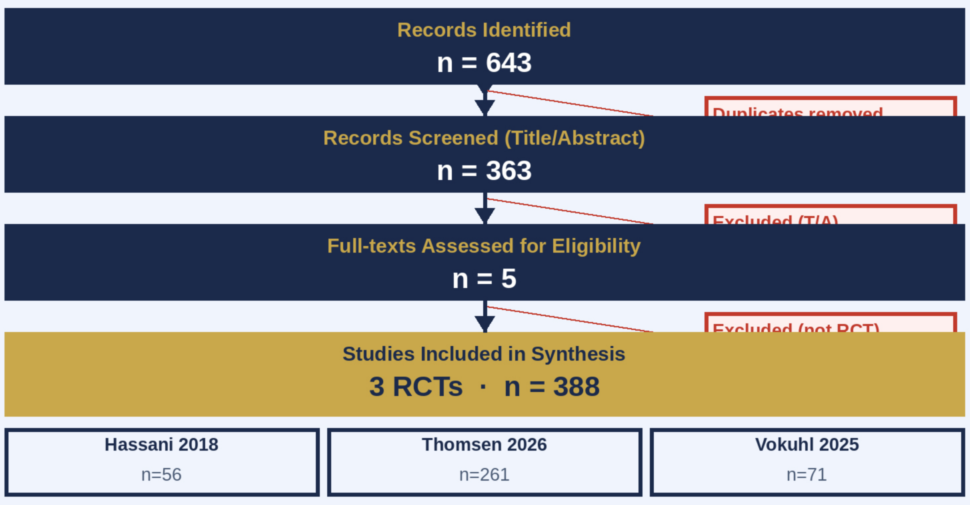
Search: "Norepinephrine" AND ("push dose" OR "bolus")
AND ("intraoperative hypotension" OR "surgery")

Inclusion: Adults ≥18 yr · Non-obstetric surgery · RCT only

Quality: Cochrane RoB 2 tool
Certainty: GRADE framework

Outcomes: MAP reversal · AUC MAP<65 · ARV-MAP
Safety: HTN overshoot · Arrhythmia · Extravasation

PRISMA 2020 FLOW DIAGRAM



RESULTS

3 RCTs · n=388 · NE 10 mcg bolus · Peripheral IV (10 mcg/mL)

► vs. Ephedrine (Hassani 2018):
NE superior MAP (98.9 vs 91.5 mmHg; p<0.001);
fewer hypotension episodes (p=0.005)

► vs. Continuous NE — Low–Mod Risk (Thomsen 2026):
Comparable AUC MAP<65 (p=0.070); 66% lower dose (p<0.001)

► vs. Continuous NE — High Risk (Vokuhl 2025):
Continuous infusion better ARV-MAP (p<0.001, d=1.00)

► Safety: No HTN overshoot or arrhythmia; zero extravasation

GRADE: MAP reversal □□○○ Low | HTN overshoot □□○○ Low
Time to target □○○○ Very Low | Arrhythmia – No evidence

RISK OF BIAS (Cochrane RoB 2)

	D1 Randomis.	D2 Deviasi	D3 Missing	D4 Outcome	D5 Sel.Rep.	Overall
Hassani 2018	!	□	□	□	□	!
Thomsen 2026	□	!	□	□	□	!
Vokuhl 2025	□	!	□	□	□	!

Low risk Some concerns (inherent to open-label design)

SUMMARY OF INCLUDED STUDIES

Study	Population	Intervention vs Comparator	Key Findings
Hassani 2018 Iran · n=56	**Hypertensive GA** spinal surgery	**NE 10mcg bolus vs** Ephedrine 5mg bolus	**NE superior MAP (p<0.001)** Fewer doses, less bradycardia
Thomsen 2026 Denmark · n=261	**ASA≥2 low-mod** noncardiac surg.	**NE Bolus vs** Continuous NE infusion	**AUC MAP<65 comparable (p=0.070)** 66% lower dose (p<0.001)
Vokuhl 2025 Germany · n=71	**ASA II-IV high-risk** elective major surg.	**NE Bolus vs** Continuous NE infusion	**Infusion better ARV-MAP (p<0.001,d=1.00)** Peripheral IV — zero extravasation

CONCLUSIONS

Push-dose NE 10 mcg/bolus via peripheral IV is effective and safe for IOH.

□ Superior to ephedrine in MAP restoration
□ Comparable to continuous infusion in low–mod risk (66% drug saving)
□ Continuous infusion preferred in high-risk patients with arterial line

GRADE: Low (□□○○) — Multicenter RCTs with standardised outcomes needed

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